Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/15/2025
PI Name:	Hein, Travis Ph.D.		
Project Title:	Temporal Impact of Simulated Microgravity on Ocular Vascular Hydrodynamics		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	thein@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979-436-9261
Organization Name:	Texas A&M University		
PI Address 1:	Medical Research & Education Building II		
PI Address 2:	8447 Riverside Parkway		
PI Web Page:			
City:	Bryan	**State:**	TX
Zip Code:	77807	**Congressional District:**	17
Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
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Key Personnel Changes/Previous PI:	NONE		
COI Name (Institution):	Zawieja, David Ph.D. (Texas A&M University System) Tharakan, Binu Ph.D. (Morehouse School of Medicine) Macias, Brandon Ph.D. (NASA Johnson Space Center) Lee, Stuart Ph.D. (KBR/NASA Johnson Space Center) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center)		
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Performance Goal Text:	
Task Description:	Spaceflight associated neuro-ocular syndrome (SANS) is reported to affect about 40% of astronauts completing long-duration spaceflights and is characterized as the development of one or more findings: optic disc edema, hyperopic shifts, globe flattening, cotton-wool spots, retinal nerve fiber layer thickening, or choroidal folds. Although prolonged exposure to the headward fluid shift occurring in microgravity is regarded as the primary instigating factor, the pathophysiology of SANS remains unclear. An additional gap in our knowledge is whether ground-based models of microgravity can simulate SANS. Therefore, ground-based studies of ocular vascular hydrodynamics are required to clarify if chronic mild elevations of ocular pressure variables compromise ocular structure and function. Since all blood and lymph vessels are compliant, fluid-filled structures whose pressures are strongly influenced by gravity, we propose to focus on potential changes directly to the ocular vasculature caused by simulated microgravity. Perfusion of the optic nerve and inner retina for sufficient delivery of oxygen is dependent on retinal blood flow. Recent studies have reported elevated central retinal artery blood flow velocity during long-term spaceflight. A sustained increase in retinal blood flow during microgravity exposure may contribute to optic disc edema because of the elevated transcapillary pressure. In addition, a microgravity induced headward venous fluid shift, resulting in elevated cephalad venous pressures, may impede removal of ocular interstitial fluid, resulting in optic disc edema and contribute to SANS. The development of optic disc edema also may be exacerbated by disruption of the blood-retinal barrier and/or reduction in ocular lymph flow. A potential molecular mechanism promoting these vascular changes in the eye is an inflammatory response related to edema formation. Elevated tissue fluid increases the diffusion distance for oxygen and other nutrients. Edema also compromises diffusion mediated removal of toxic cellular metabolites. However, so far there has been no systematic analysis of ocular vascular changes in microgravity that may promote ocular edema and/or other characteristics of SANS. To address these information gaps, we have assembled a team of experts in SANS and all 3 main vascular types (arteries, veins, and lymphatics). The objective of this application is to determine whether the hindlimb unloading (HU) model of simulated microgravity causes a time-dependent change in structure and function of the ocular vasculature at the level of feed arteries, venous exchange vessels and lymph vessels, and whether sex differences influence these responses. We will perform in vivo measures of vascular function (retinal artery blood flow velocity, retinal arteriolar and venular diameters, and retinal venular permeability) and in vitro studies of isolated vessel structure and function (vessel histology, arterial vasomotor regulation, and lymphatic transport characteristics) in male and female mice after 2- and 4-wk of HU. The central hypothesis of this proposal is that simulated microgravity causes temporal changes in structure/function of the ocular vasculature and subsequent alterations in ocular hydrodynamics leading to SANS. We will pursue 3 specific aims: (1) Determine the temporal impact of simulated microgravity on ocular artery structure/function; (2) Determine the temporal impact of simulated microgravity on ocular vein structure/function; (3) Determine the temporal impact of simulated microgravity on ocular lymphatic structure/function. Our team approach to accomplish these studies will provide a novel comprehensive evaluation of the temporal impact of simulated microgravity on ocular hydrodynamics in mice, and correlation of these findings with SANS. Outcomes will help identify the strengths and limitations of a murine HU model for SANS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our ground-based studies in this project will address whether relatively short- and long-term simulated microgravity using the hindlimb unloading model in mice can simulate signs and symptoms of spaceflight associated neuro-ocular syndrome (SANS), such as ocular edema and thickening of the retinal tissue. We will also determine whether simulated microgravity causes a time-dependent change in structure and function of the blood vessels in the eye, and whether ocular vascular changes correlate with SANS. Results from this project are expected to provide the first insight into the impact of simulated microgravity on vascular hydrodynamics in the eye. The expected findings will likely also provide new insight into vascular complications relevant to ocular diseases in humans on Earth, such as glaucoma and ocular hypertension.
Task Progress:	For Aim 1A, we completed the total retinal thickness (TRT) measurements from the internal limiting membrane (ILM) to the retinal pigment epithelium (RPE) in the retinal images obtained by optical coherence tomography (OCT). The TRT change within 200 μm of the optic disc in male and female sham and hindlimb unloading (HU) mice after 2 and 4 wk was measured. There is a trend of elevated TRT in female and male mice after 4 wk but not 2 wk of HU, but this is not statistically significant. We performed the OCT in these mice after HU with the mice no longer in the HU position. To make sure the reloading of the mice after HU was not influencing the data, we have developed an approach to allow for maintaining the HU position during the OCT data collection. We have begun these experiments and should be able to complete them within 2-3 months and then the data analysis within one month. We will be able to complete this data collection and analysis in male and female mice during a no-cost extension of 6 months. For Aim 2A, the proposed studies to assess retinal microvascular permeability in HU and Sham control mice were completed. Mice received FITC-dextran (10-kDA) as a fluorescent tracer for intravital imaging of the retinal microcirculation. The data show that there is no significant change in retinal microvascular permeability between the 4-wk HU and 4-wk sham control male mice. For Aim 2A, we also completed a behavioral study to test for potential near vision changes in the male 4-wk HU and 4-wk sham mice, using a visual cliff test. The visual cliff test is used for the depth perception and the visualization of perceived barriers. The results demonstrate no significant change in near vision in male HU mice compared to sham.
Bibliography Type:	Description: (Last Updated: 07/07/2026)
Abstracts for Journals and Proceedings	Hightower G, Pham D, Wang W, Babu S, Olson B, Hoppe J, Laurie S, Macias B, Tharakan B, Zawieja DC, Hein TW. "Temporal impact of simulated microgravity on ocular vascular hydrodynamics in male and female mice." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Poster. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025

Articles in Peer-reviewed Journals	Waithe OY, Anderson A, Muthusamy S, Seplovich GM, Tharakan B. "Homocysteine induces brain and retinal microvascular endothelial cell barrier damage and hyperpermeability via NLRP3 inflammasome pathway differentially." Microcirculation. 2025 Jul 17;32(5):e70019. https://doi.org/10.1111/micc.70019 ; PMID: 40674175 , Jul-2025